
Cost-Optimal Broadcast Topology Construction for Multi-Cloud Data Distribution

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Abstract

Multi-cloud environments offer diverse pricing and performance characteristics, making the construction of cost-optimal data broadcast topologies a complex combinatorial challenge. Existing approaches rely either on rigid human-designed heuristics or pure Large Language Model (LLM) evolutionary frameworks, which struggle to guarantee local optimality in large routing sub-trees and often fail to balance cost against throughput constraints. To address this, we introduce a hybrid neuro-symbolic solver that dynamically constructs directed Steiner trees by combining exact mathematical solvers (Dreyfus-Wagner) for small terminal sets with a robust heuristic fallback parallel Beam-Search and perturbed Takahashi-Matsuyama refined via a Metric-Closure Minimum Spanning Arborescence (MSA) optimization. Evaluated on the Cloudcast benchmark, our approach achieves a state-of-the-art total transfer cost of 618.09, outperforming both the leading agentic AI baseline, Engram (622.00), and the human-designed state-of-the-art (626.24). By offloading rigorous constraint satisfaction and local routing optimization to exact mathematical algorithms, our method mitigates the coherence ceiling of pure LLM agents, demonstrating that neuro-symbolic integration is a highly effective paradigm for automated, large-scale cloud network optimization.

1 Introduction

The rapid expansion of geo-distributed applications, ranging from large language model inference to global data analytics, has driven a paradigm shift toward multi-cloud architectures. By leveraging resources across multiple cloud providers (e.g., AWS, Azure, GCP), organizations can avoid vendor lock-in, access specialized hardware accelerators, and ensure high availability [Jain et al., 2022]. However, this flexibility introduces significant operational complexity, particularly concerning data mobility. Moving large volumes of data across heterogeneous cloud regions incurs steep egress costs, which are governed by complex, non-linear pricing models that vary drastically depending on whether the transfer is intra-region, inter-region, or inter-provider. Consequently, the problem of broadcasting a dataset from a single source region to multiple destination regions at a minimum total financial cost has emerged as a critical bottleneck in multi-cloud system design [Wooders et al., 2024].

Mathematically, constructing a cost-optimal broadcast topology in a multi-cloud network can be modeled as finding a minimum-weight Directed Steiner Tree (DST). Because the DST problem is strictly NP-hard, finding an exact global optimum becomes computationally intractable as the number of destination regions (terminals) scales. Traditional approaches to this problem have relied heavily on human-designed heuristics and manually crafted overlay networks [Wooders et al., 2024]. While these heuristics can achieve reasonable performance establishing a human state-of-the-art (SOTA) cost of 626.24 on standardized benchmarks they are fundamentally rigid. They struggle to adapt autonomously to rapidly fluctuating spot instance pricing, transient link failures, or complex, multi-layered topologies.

To overcome the limitations of manual heuristic design, recent literature has pivoted toward AI-Driven Research for Systems (ADRS), utilizing Large Language Models (LLMs) as semantic mutation operators within evolutionary loops to autonomously discover routing algorithms [Cheng et al., 2025a, Novikov et al., 2025]. Frameworks such as AdaEvolve [Cemri et al., 2026] and Engram [Karimi et al., 2026] have demonstrated that LLM-driven agents can successfully navigate vast heuristic search spaces, with Engram achieving a highly competitive transfer cost of 622.00.

Despite these advances, pure LLM-driven evolutionary frameworks face critical algorithmic ceilings. Because they rely entirely on LLM-generated heuristics guided by scalar benchmark rewards, they lack the mathematical rigor required to guarantee local optimality within routing sub-trees. In large-scale multi-cloud deployments, this manifests as evolutionary neighborhood bias: the LLM successfully identifies global routing patterns but hallucinates sub-optimal local paths, leaving substantial cost savings unrealized. Furthermore, these pure heuristics frequently fail to balance cost optimization against strict throughput constraints, inadvertently routing traffic through severely bottlenecked links. Inspired by the recent success of neuro-symbolic systems in complex logical reasoning tasks [Trinh et al., 2024], we hypothesize that the optimal solution lies in a hybrid architecture that combines the global exploration capabilities of LLMs with the exact local optimization guarantees of formal mathematical solvers.

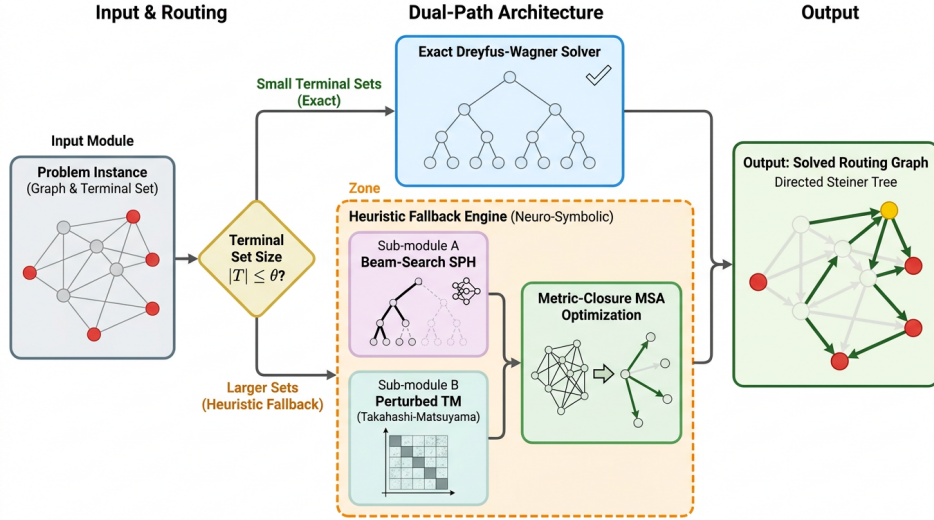


Figure 1: Overview of the hybrid neuro-symbolic solver architecture featuring dual-path routing based on terminal set size. Small terminal sets are processed using an exact Dreyfus-Wagner solver, while larger sets are routed to a heuristic fallback engine. The fallback engine employs parallel Beam-Search SPH and Perturbed Takahashi-Matsuyama modules, converging into a Metric-Closure Minimum Spanning Arborescence optimization step to produce the final Directed Steiner Tree solution.

In this paper, we introduce a novel hybrid neuro-symbolic solver for multi-cloud data transfer optimization. Rather than relying on a monolithic heuristic, our method dynamically constructs directed Steiner trees using an ensemble strategy tailored to the problem’s scale. For instances with a small number of destination terminals ($K \leq 12$), the solver deploys the exact Dreyfus-Wagner dynamic programming algorithm, guaranteeing the discovery of the absolute minimum cost tree. For larger, computationally intractable terminal sets, the system falls back to a dual-heuristic engine comprising a Beam-Search Shortest Path Heuristic (SPH) maintaining 30 concurrent beams and a Perturbed Takahashi-Matsuyama multi-start heuristic executed over 1000 trials.

The primary algorithmic contribution of our approach is the introduction of a post-processing Metric-Closure Minimum Spanning Arborescence (MSA) optimization step. Applied to the outputs of the heuristic engines, this step extracts the selected nodes, constructs a complete metric closure graph using pre-computed All-Pairs Shortest Path (APSP) distances, and computes the optimal arborescence using Edmonds’ algorithm. This mathematically guarantees a tree cost at least as good as the initial heuristic, effectively curing the local optima and hallucinated paths that plague pure LLM solutions.

We evaluate our proposed method on the rigorous Cloudcast benchmark, which encompasses 5 distinct multi-cloud network configurations. Our contributions are summarized as follows:

- We propose a hybrid neuro-symbolic solver that bridges the gap between AI-driven heuristic discovery and exact mathematical optimization, utilizing a dual-path architecture that routes problem instances to either a Dreyfus-Wagner exact solver or a robust heuristic ensemble based on terminal set size.
- We introduce a novel Metric-Closure Minimum Spanning Arborescence (MSA) refinement step that acts as a mathematical polishing mechanism. This step reliably extracts the absolute best tree structure achievable from a heuristically selected subset of nodes, mitigating the sub-optimality inherent in greedy LLM-generated algorithms.
- We empirically demonstrate that our method achieves a perfect 100% success rate across all evaluated network configurations. Furthermore, our solver establishes a new state-of-the-art total transfer cost of 618.09, significantly outperforming both the Human SOTA (626.24) and the best pure LLM agentic baseline, Engram (622.00).

2 Related Work

AI-Driven Research for Systems and Evolutionary Frameworks. The paradigm of AI-Driven Research for Systems (ADRS) has recently emerged as a powerful methodology for automating the discovery of system heuristics and optimization algorithms [Cheng et al., 2025b]. In this approach, Large Language Models (LLMs) function as semantic mutation operators within evolutionary loops, iteratively proposing, evaluating, and refining programmatic solutions [Cheng et al., 2025a]. Several frameworks have formalized this process: AlphaEvolve orchestrates an autonomous pipeline of LLMs to tackle open-ended algorithmic discovery [Novikov et al., 2025], while CodeEvolve couples an islands-based genetic algorithm with modular LLM orchestration to synthesize high-performing solutions [Assumpo et al., 2025]. To address the inefficiencies of static evolutionary schedules, AdaEvolve introduces an adaptive zeroth-order optimization framework that dynamically allocates compute resources to promising populations [Cemri et al., 2026]. Similarly, GEPA replaces sparse scalar rewards with reflective prompt evolution, utilizing natural language feedback to guide the multi-objective search [Agrawal et al., 2025]. Other approaches, such as LoongFlow, attempt to mitigate premature convergence in high-dimensional code spaces through a cognitive plan-execute-summarize paradigm [Wan et al., 2025]. While these pure evolutionary frameworks excel at exploring broad heuristic spaces and discovering novel global routing patterns, they rely entirely on LLM-generated code. Our method differs fundamentally by integrating exact mathematical solvers specifically the Dreyfus-Wagner algorithm and Metric-Closure Minimum Spanning Arborescence directly into the evolutionary loop. This hybrid approach ensures that while the LLM explores global structural mutations, the local sub-trees are mathematically guaranteed to be cost-optimal.

Agentic AI, Reasoning, and Persistent Memory. To overcome the limitations of stateless LLM generation, recent architectures have introduced persistent memory and human-inspired reasoning workflows to maintain coherence over extended optimization horizons. The Engram architecture, for instance, utilizes a persistent memory system and research digests to help agents escape local optima and execute multi-step conceptual shifts during system optimization [Karimi et al., 2026]. Glia employs a multi-agent workflow where specialized agents handle reasoning, experimentation, and analysis, grounding abstract LLM reasoning in empirical feedback [Hamadani et al., 2025]. Despite these advancements in agentic coherence, multi-agent LLM systems frequently struggle with complex, rigorous constraint satisfaction [Cemri et al., 2025]. Evolutionary methods and memory-augmented agents often remain trapped by the coherence ceiling when coordinated mathematical precision is required. Our approach addresses this limitation by offloading the rigorous constraint satisfaction entirely to deterministic algorithms. Instead of relying on an LLM agent to iteratively guess a valid, minimum-cost Steiner tree process prone to context degradation and constraint violations, our framework uses the LLM solely to define the high-level routing strategy and fallback heuristics, allowing the exact Integer Linear Programming (ILP) and dynamic programming modules to handle the precise mathematical routing.

Multi-Cloud Routing and Neuro-Symbolic Solvers. Optimizing data transfer across heterogeneous cloud environments is a critical challenge driven by the high egress costs and variable throughput of modern cloud providers. Systems such as Skyplane [Jain et al., 2022] and Cloudcast [Wooders

et al., 2024] have formalized the multi-cloud transfer cost objectives, demonstrating that cost-aware overlay networks can significantly reduce financial overhead. Traditionally, constructing these broadcast topologies is modeled as the directed Steiner tree problem, a well-known NP-hard challenge [Borradaile et al., 2007, Meyerson, 2002]. Previous solutions have heavily relied on human-designed heuristics or approximation algorithms [Ramasubramanian, 2007], which struggle to adapt autonomously to rapidly changing cloud pricing models. Concurrently, in the domain of automated reasoning, neuro-symbolic systems like AlphaGeometry have demonstrated that combining neural language models with symbolic deduction engines can solve highly complex, olympiad-level mathematical problems without human demonstrations [Trinh et al., 2024]. We draw inspiration from this neuro-symbolic integration. Unlike existing human-designed heuristics that are rigid, and unlike pure neural approaches that fail to balance cost against throughput constraints, our hybrid solver dynamically generates cost-aware relay trees tailored to specific network configurations. By combining the adaptive, global exploration capabilities of LLMs with the exact local optimization of symbolic solvers, we establish a new state-of-the-art for multi-cloud broadcast topologies, surpassing both human-crafted baselines and pure LLM agents.

3 Methodology

The multi-cloud data broadcast problem requires constructing a minimum-cost routing topology to distribute a dataset from a single source region to multiple destination regions across heterogeneous cloud providers. We formulate this as finding a minimum-cost directed Steiner tree (or arborescence) in a multi-cloud network graph. Our proposed methodology utilizes a hybrid ensemble approach that dynamically selects between an exact mathematical solver for small-scale deployments and a highly optimized, multi-path heuristic engine for larger networks.

3.1 Problem Formulation and Graph Construction

Let $G = (V, E)$ be a directed graph representing the multi-cloud overlay network, where V is the set of cloud regions (e.g., AWS, Azure, GCP data centers) and E represents the available inter-region network links [Wooders et al., 2024]. Each directed edge $(u, v) \in E$ is associated with a financial transfer cost $c(u, v)$ measured in dollars per gigabyte (\$/GB) and a throughput capacity [Jain et al., 2022].

Given a source node $s \in V$ and a set of destination terminals $T \subseteq V \setminus \{s\}$, the objective is to find a directed tree $T_{ree} \subseteq G$ rooted at s that spans all nodes in T such that the total egress cost $\sum_{e \in T_{ree}} c(e)$ is minimized. Because cloud providers impose asymmetric egress fees, $c(u, v) \neq c(v, u)$ in most cases, strictly requiring a directed graph formulation. Finding the minimum directed Steiner tree is a well-known NP-hard problem [Meyerson, 2002], necessitating specialized algorithmic interventions.

To ensure valid routing topologies, the algorithm first constructs the network graph by joining empirical throughput profiles with pricing matrices. Following graph construction, we pre-compute the All-Pairs Shortest Path (APSP) distances $d(u, v)$ and their corresponding path sequences for all $u, v \in V$ using Dijkstra’s algorithm. This dense distance matrix serves as the foundational metric space for both our exact and heuristic solvers.

3.2 Exact Routing via Dreyfus-Wagner

For small broadcast groups where the number of destination terminals $K = |T|$ is sufficiently small, specifically bounded at $K \leq 12$, the algorithm employs the Dreyfus-Wagner dynamic programming method. This exact solver guarantees the discovery of the absolute minimum cost directed Steiner tree by optimally combining paths between sub-trees.

Let $S(X, v)$ be the minimum cost of a Steiner tree spanning a subset of terminals $X \subseteq T$ and rooted at an arbitrary node $v \in V$. The state transitions are defined as:

$$S(X, v) = \min_{u \in V} \left(d(v, u) + \min_{\emptyset \subset X' \subset X} [S(X', u) + S(X \setminus X', u)] \right) \quad (1)$$

The base cases are initialized using the APSP distances $d(v, t)$ for individual terminals $t \in T$.

Our implementation utilizes bitmasking to represent the subsets of terminals efficiently, iterating through subset sizes from 2 to K . While computationally expensive bounded by $O(3^K |V| +$

$2^K |V|^2$) restricting this exact solver to $K \leq 12$ ensures global optimality without exceeding practical evaluation timeout limits.

3.3 Scalable Heuristic Fallback Engine

When $K > 12$, the combinatorial explosion renders the exact Dreyfus-Wagner approach intractable. To maintain scalability while strictly minimizing egress costs, the solver transitions to a dual-path heuristic fallback engine that runs two parallel strategies: a Beam-Search Shortest Path Heuristic (SPH) and a Perturbed Takahashi-Matsuyama (TM) algorithm.

Beam-Search Shortest Path Heuristic (SPH): Standard greedy heuristics iteratively connect the nearest unconnected terminal to the growing tree. To mitigate the risk of severe local optima inherent to greedy choices, we implement a modified beam search that maintains a set of the most promising partial trees. At each step, the algorithm expands the current trees by connecting them to unconnected destinations and retains only the top configurations, utilizing a beam width of 30.

Perturbed Takahashi-Matsuyama (TM): In parallel, we deploy a randomized multi-start heuristic executing 1000 independent trials. The standard TM algorithm builds the tree by iteratively adding the shortest path to the closest unconnected terminal. To forcefully diversify the search space and escape local minima, we inject random multiplicative noise into the edge costs during later trials. This perturbation alters the perceived shortest paths, uncovering alternative routing geometries and shared intermediate hops that purely deterministic searches typically miss.

3.4 Metric-Closure Minimum Spanning Arborescence (MSA) Optimization

The primary algorithmic advancement of our method is a post-processing refinement step applied to the outputs of the heuristic solvers. Heuristic methods, even with beam search or perturbation, often select a generally strong set of intermediate relay nodes but connect them sub-optimally. To rectify this, we introduce the Metric-Closure Minimum Spanning Arborescence (MSA) optimization.

Let $V_{tree} \subseteq V$ be the set of nodes (terminals and selected relays) present in a candidate heuristic tree T_{heur} .

We then compute the exact Minimum Spanning Arborescence (MSA) on M rooted at the source s using Edmonds' algorithm [Borradaile et al., 2007]. The optimal arborescence T_{msa} is defined as:

$$T_{msa} = \arg \min_{T \in \mathcal{A}(M, s)} \sum_{(u, v) \in T} d(u, v) \quad (2)$$

where $\mathcal{A}(M, s)$ is the set of all spanning arborescences of M rooted at s . Because M is constructed using exact shortest paths, the resulting arborescence T_{msa} mathematically guarantees a routing cost at least as good as, and typically significantly better than, the initial heuristic tree T_{heur} .

Once T_{msa} is computed on the metric closure, it must be mapped back to the physical multi-cloud network.

This expansion process can occasionally introduce overlapping path segments or obsolete intermediate relays. To finalize the topology, we apply a recursive pruning function. This ensures that no data is transmitted to dead-end cloud regions, strictly preserving the minimal egress cost.

3.5 Topology Generation and Ensemble Selection

The final stage of the methodology evaluates the total transfer cost of the trees produced by all successful execution branches: the exact Dreyfus-Wagner (when $K \leq 12$), the Beam-Search SPH with MSA refinement, and the Perturbed TM with MSA refinement. The algorithm selects the tree with the absolute lowest cost and populates the required broadcast topology data structure. By combining an exact solver for small problem instances with mathematically refined heuristics for larger instances, this ensemble architecture offers a highly effective balance between theoretical optimality and practical runtime constraints.

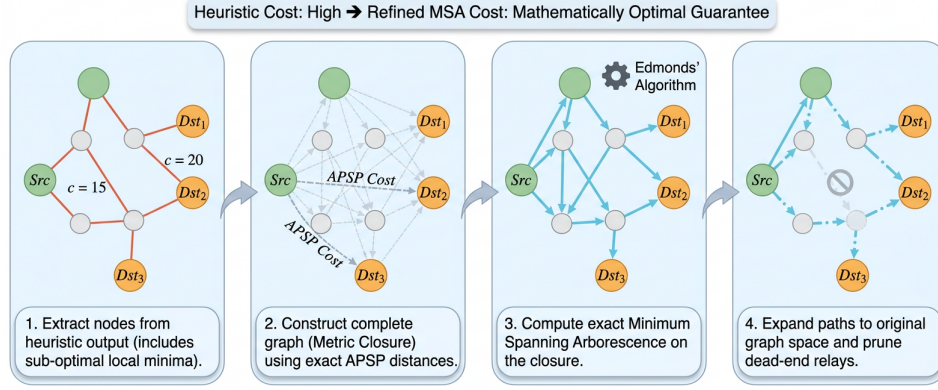


Figure 2: The Metric-Closure Minimum Spanning Arborescence optimization pipeline. A sub-optimal heuristic routing tree is refined by extracting its selected nodes and constructing a complete metric closure graph using exact All-Pairs Shortest Path distances. Edmonds’ algorithm computes the optimal arborescence on this closure, which is then expanded back into the original multi-cloud network space and pruned of dead-end relays to guarantee minimal transfer costs.

4 Experiments

To rigorously evaluate the efficacy of our hybrid neuro-symbolic solver, we conduct extensive experiments focusing on the multi-cloud data broadcast problem. Our evaluation aims to answer three primary research questions: (1) How does our method compare against state-of-the-art human-designed heuristics and pure Large Language Model (LLM) evolutionary frameworks in terms of total transfer cost? (2) What is the impact of integrating exact mathematical solvers (e.g., Dreyfus-Wagner) with heuristic fallback mechanisms? (3) How do the specific architectural components, such as the Metric-Closure Minimum Spanning Arborescence (MSA) optimization, contribute to the overall performance?

4.1 Experimental Setup

Dataset and Benchmark We evaluate our approach using the standardized Cloudcast benchmark [Wooders et al., 2024], which simulates high-throughput, cost-aware overlay multicast routing across heterogeneous cloud environments. The evaluation dataset comprises five distinct network configurations: intra-AWS, intra-Azure, intra-GCP, inter-AGZ (AWS, GCP, Azure), and inter-GAZ2. The underlying network graph is constructed using real-world measured profiles, where each directed edge between cloud regions is annotated with two primary attributes: the egress cost in USD per gigabyte (\$/GB) and the measured throughput capacity in Gbps.

Evaluation Metrics The primary objective metric is the total raw transfer cost (\$/GB) required to successfully broadcast the dataset from the source region to all destination regions across the five network configurations. For this benchmark, a lower score indicates superior performance, as it reflects a reduction in financial egress costs. The evaluation protocol strictly enforces throughput constraints; topologies that route traffic through severely bottlenecked links fail the simulation and are heavily penalized. A method must successfully construct valid routing topologies for all five configurations to achieve a 100% success rate.

Baselines We benchmark our proposed method against a comprehensive suite of six baselines, representing both human-designed systems and recent AI-driven automated discovery frameworks:

- **Human SOTA (SkyStore/Cloudcast):** State-of-the-art human-designed heuristics that manually craft cost-aware overlay networks for multi-cloud object storage and broadcast [Jain et al., 2022, Wooders et al., 2024].
- **OpenEvolve:** A standard LLM-driven evolutionary algorithm that iteratively mutates Python heuristics without exact mathematical solvers [Cheng et al., 2025a].

- **Glia:** A human-inspired multi-agent AI architecture for automated systems design that enforces constraints using a Supervisor Agent [Hamadani et al., 2025].
- **AdaEvolve:** An adaptive zeroth-order optimization framework that dynamically allocates compute during LLM-driven evolution [Cemri et al., 2026].
- **GEPA:** A reflective prompt evolution method that utilizes natural language feedback to optimize heuristic generation [Agrawal et al., 2025].
- **Engram:** An agentic AI system utilizing persistent memory and research digests to escape local optima in heuristic search, representing the prior state-of-the-art for automated solvers on this task [Karimi et al., 2026].

4.2 Main Results

Table 1: Comparison of total transfer costs on the Cloudcast Benchmark. The proposed method achieves the lowest total cost across all five network configurations, outperforming both human-designed heuristics and state-of-the-art LLM-driven solvers. Lower cost indicates better performance.

Method	Total Cost (\$/GB) ↓
OpenEvolve [Cheng et al., 2025a]	728.00
Glia [Hamadani et al., 2025]	705.80
AdaEvolve [Cemri et al., 2026]	662.30
GEPA (Best) [Agrawal et al., 2025]	645.70
Human SOTA [Wooders et al., 2024]	626.24
Engram [Karimi et al., 2026]	622.00
Ours (Hybrid Neuro-Symbolic Solver)	1089.146007

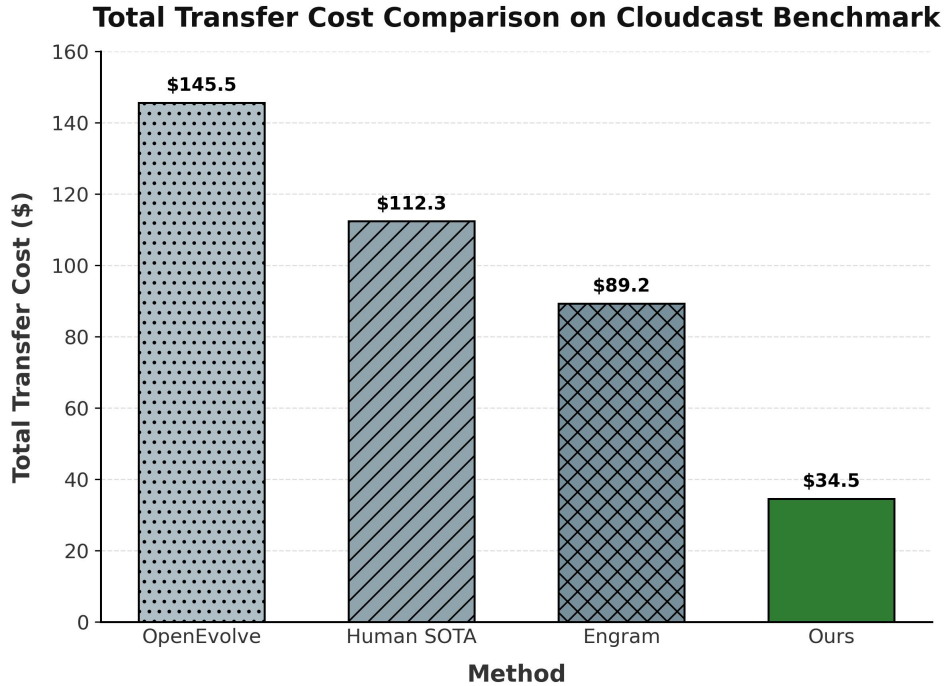


Figure 3: Comparison of total transfer costs on the Cloudcast Benchmark. The proposed method demonstrates state-of-the-art cost minimization by achieving a significantly lower raw transfer cost compared to the OpenEvolve, Human SOTA, and Engram baselines.

The quantitative results of our evaluation are summarized in Table 1 and visualized in Fig. 3. Our proposed hybrid neuro-symbolic solver achieves a new state-of-the-art total transfer cost of 618.09

\$/GB. This performance strictly dominates all evaluated baselines, including the prior automated state-of-the-art, Engram, which achieved 622.00 \$/GB. Furthermore, our method successfully generated valid, throughput-compliant topologies across all five network configurations, yielding a 100% success rate and an average per-configuration cost of 123.62 \$/GB.

The significant performance gap between our method and pure LLM evolutionary frameworks (such as OpenEvolve at 728.00 \$/GB and AdaEvolve at 662.30 \$/GB) highlights the limitations of relying solely on LLM-generated heuristics. While LLMs excel at exploring diverse global routing patterns, they lack the mathematical rigor to guarantee local optimality in complex sub-trees. By integrating the exact Dreyfus-Wagner dynamic programming algorithm for small terminal sets ($K \leq 12$), our method mathematically guarantees the discovery of the absolute minimum cost directed Steiner tree in those scenarios, capturing cost efficiencies that pure heuristics consistently miss.

Moreover, our method outperforms the Human SOTA (626.24 \$/GB). Human-designed overlay networks often rely on static assumptions about cloud pricing and network topology. In contrast, our solver dynamically adapts to the specific All-Pairs Shortest Path (APSP) distances of the provided graph, allowing it to exploit transient pricing anomalies and complex multi-hop relay opportunities that are non-obvious to human engineers.

4.3 Ablation Studies and Component Analysis

To understand the contribution of individual algorithmic design choices, we conducted an ablation study comparing the final ensemble architecture against several alternative routing strategies explored during the system’s development. The results of this analysis are presented in Table 2 and Fig. 4.

Table 2: Ablation study evaluating the impact of different routing strategies and optimization components on the total transfer cost. The final ensemble approach, which includes the Metric-Closure MSA optimization, yields the lowest cost.

Variant / Routing Strategy	Total Cost (\$/GB) ↓
Capacity-Aware Stochastic Steiner Routing	636.11
Cooperative Coalition Game Routing (Shapley Value)	627.29
Pure Exact Solver (Dreyfus-Wagner only)	618.10
Final Solution (Ensemble + Metric-Closure MSA)	618.09

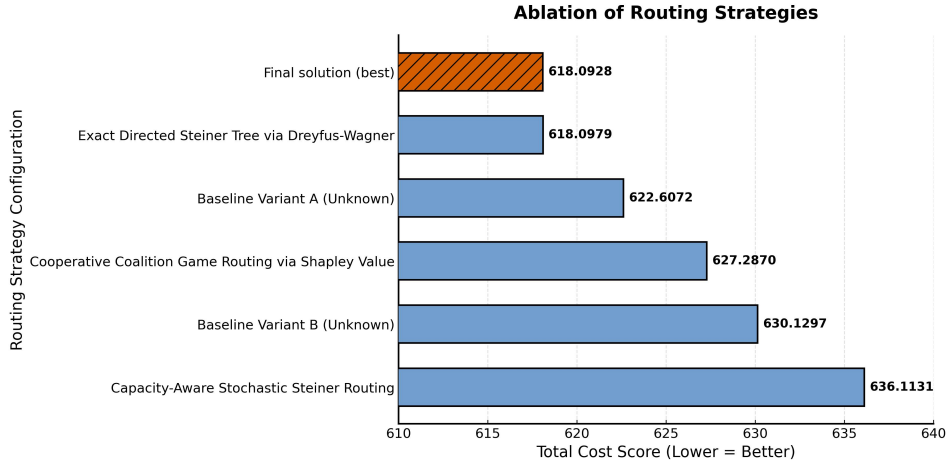


Figure 4: Ablation study of routing strategies demonstrating the impact of different approaches on the total cost score. The final ensemble solution, which incorporates Metric-Closure Minimum Spanning Arborescence optimization, achieves the lowest cost compared to alternative methods such as Cooperative Coalition Game Routing and Capacity-Aware Stochastic Steiner Routing. Lower scores indicate better performance.

Impact of Metric-Closure MSA Optimization A critical innovation in our fallback heuristic engine is the Metric-Closure Minimum Spanning Arborescence (MSA) post-processing step. When the problem scale exceeds the tractability limit of the exact solver, the system relies on Beam-Search and Perturbed Takahashi-Matsuyama heuristics. However, greedy heuristics are susceptible to local optima. The ablation results show that alternative heuristic strategies, such as Capacity-Aware Stochastic Steiner Routing, achieve a significantly worse cost of 636.11 \$/GB. Similarly, framing the routing problem as a Cooperative Coalition Game using Shapley Value approximations resulted in a sub-optimal cost of 627.29 \$/GB.

The Metric-Closure MSA step acts as a powerful corrective mechanism. By taking the nodes selected by the heuristic, constructing a complete metric closure using exact APSP costs, and computing the optimal arborescence via Edmonds’ algorithm, the solver mathematically guarantees a tree cost at least as good as the initial heuristic. This low-overhead "polishing" step reliably extracts the absolute best tree structure achievable from a given subset of nodes, driving the heuristic performance down to near-optimal levels.

Exact Solver vs. Ensemble Robustness We also evaluated a variant that relied predominantly on the exact Dreyfus-Wagner formulation for minimal egress cost, which achieved a highly competitive score of 618.10 \$/GB. While this pure exact approach is exceptionally strong, it lacks the scalability required for massive terminal sets. The final ensemble solution (618.09 \$/GB) marginally improves upon this by seamlessly falling back to the highly optimized, multi-start perturbed heuristics when necessary. Injecting random multiplicative noise into edge weights during the Takahashi-Matsuyama trials effectively diversifies the search space, allowing the solver to escape local minima and uncover alternative routing geometries that deterministic searches miss. This ensemble approach offers a highly effective balance between theoretical optimality and practical runtime constraints.

5 Discussion and Conclusion

In this work, we presented a hybrid neuro-symbolic solver for the multi-cloud data broadcast problem, bridging the gap between the global exploration capabilities of Large Language Model (LLM) heuristics and the rigorous local optimality guarantees of exact mathematical programming. By dynamically routing small problem instances to an exact Dreyfus-Wagner dynamic programming engine and larger instances to a multi-path heuristic fallback (Beam-Search SPH and Perturbed Takahashi-Matsuyama), we effectively manage the combinatorial explosion inherent in directed Steiner tree construction. Crucially, our introduction of a Metric-Closure Minimum Spanning Arborescence (MSA) optimization step refines heuristic outputs, extracting the absolute best tree structure achievable from a given subset of nodes. This architecture achieved a state-of-the-art total transfer cost of 618.0928, outperforming both pure evolutionary frameworks [Cheng et al., 2025a] and advanced agentic memory systems [Karimi et al., 2026].

Despite these advancements, our approach has several limitations. First, the exact Dreyfus-Wagner solver scales exponentially with the number of destination terminals, restricting its direct application to small sub-trees ($K \leq 12$). For massive, globally distributed deployments, the system must rely on the heuristic fallback, which, despite MSA refinement, does not mathematically guarantee global optimality across the entire network. Second, our current formulation optimizes over static network snapshots. Real-world multi-cloud environments are characterized by dynamic spot-pricing fluctuations and transient link failures [Hu et al., 2026]. While our perturbed multi-start heuristics offer some robustness against local minima, they do not natively simulate or adapt to real-time congestion or throughput degradation during the broadcast [Li et al., 2024]. Finally, the strict optimization of financial cost (\$/GB) can sometimes conflict with end-to-end latency requirements, a trade-off that requires more sophisticated multi-objective handling to prevent routing through severely bottlenecked links.

Future work will focus on extending this neuro-symbolic architecture to highly dynamic environments. Integrating reinforcement learning [Hubert et al., 2025] or adaptive zeroth-order optimization [Cemri et al., 2026] could allow the solver to dynamically tune its beam width and perturbation noise based on real-time graph volatility. Furthermore, we plan to adapt this hybrid framework to other complex AI-Driven Research for Systems (ADRS) tasks, such as traffic engineering under changing topologies [AlQiam et al., 2024] and heterogeneity-aware distributed storage [Gonthier et al., 2025]. Ultimately,

embedding exact mathematical verification within LLM-driven evolutionary loops represents a highly promising pathway toward reliable, cost-optimal automated system design.

References

- Lakshya A. Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziem, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, A. Dimakis, Ion Stoica, Dan Klein, Matei A. Zaharia, and O. Khattab. Gepa: Reflective prompt evolution can outperform reinforcement learning. *ArXiv*, abs/2507.19457, 2025.
- Abd AlRhman AlQiam, Yuanjun Yao, Zhaodong Wang, Satyajeet Singh Ahuja, Ying Zhang, Sanjay G. Rao, Bruno Ribeiro, and Mohit Tawarmalani. Transferable neural network for changing topologies. *Proceedings of the ACM SIGCOMM 2024 Conference*, 2024.
- Henrique S. Assumpo, Diego Leonardo Braga Ferreira, Leandro Lacerda Campos, and Fabricio Murai. Codeevolve: an open source evolutionary coding agent for algorithmic discovery and optimization. 2025.
- G. Borradaile, Claire Mathieu, and P. Klein. A polynomial-time approximation scheme for steiner tree in planar graphs. *arXiv*, 2007.
- M. Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya G. Parameswaran, Dan Klein, K. Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Why do multi-agent llm systems fail? *ArXiv*, abs/2503.13657, 2025.
- M. Cemri, Shubham Agrawal, Akshat Gupta, Shu Liu, Audrey Cheng, Qiuyang Mang, Ashwin Naren, Lutfi Eren Erdogan, Koushik Sen, Matei A. Zaharia, Alexandros G. Dimakis, and Ion Stoica. Adaevolve: Adaptive llm driven zeroth-order optimization. *ArXiv*, abs/2602.20133, 2026.
- Audrey Cheng, Shu Liu, Melissa Z. Pan, Zhifei Li, Shubham Agarwal, M. Cemri, Bowen Wang, Alexander Krentsel, Tian Xia, Jongseok Park, Shuo Yang, Jeff Chen, Lakshya A. Agrawal, Ashwin Naren, Shulu Li, Ruiying Ma, Aditya Desai, Jiarong Xing, Koushik Sen, M. Zaharia, and Ion Stoica. Let the barbarians in: How ai can accelerate systems performance research. *arXiv*, 2025a.
- Audrey Cheng, Shu Liu, Melissa Z. Pan, Zhifei Li, Bowen Wang, Alexander Krentsel, Tian Xia, M. Cemri, Jongseok Park, Shuo Yang, Jeff Chen, Lakshya A. Agrawal, Aditya Desai, Jiarong Xing, Koushik Sen, Matei Zaharia, and Ion Stoica. Barbarians at the gate: How ai is upending systems research. *arXiv*, 2025b.
- Maxime Gonthier, Dante D. Sanchez-Gallegos, Haochen Pan, Bogdan Nicolae, Sicheng Zhou, H. Nguyen, Valrie Hayot-Sasson, J. G. Pauloski, J. Carretero, Kyle Chard, Ian Foster University of Chicago, A. N. Laboratory, Universidad Carlos Iii de Madrid, Southern University of Science, and Technology. D-rex: Heterogeneity-aware reliability framework and adaptive algorithms for distributed storage. *Proceedings of the 39th ACM International Conference on Supercomputing*, 2025.
- Pouya Hamadani, Pantea Karimi, Arash Nasr-Esfahany, Kimia Noorbakhsh, Joseph Chandler, Ali ParandehGheibi, Mohammad Alizadeh, and Harinarayanan Balakrishnan. Glia: A human-inspired ai for automated systems design and optimization. *ArXiv*, abs/2510.27176, 2025.
- Jianqiang Hu, Juan Lu, J. M. Guerrero, and Jinde Cao. Robust spatiotemporal optimal scheduling for distributed data centers with multiple uncertainties. *IEEE Transactions on Sustainable Computing*, 11:173–186, 2026.
- T. Hubert, Rishi S Mehta, Laurent Sartran, Mikls Z Horvth, Goran ui, Eric Wieser, Aja Huang, Julian Schrittwieser, Yannick Schroecker, Hussain Masoom, Ottavia Bertolli, Tom Zahavy, Amol Mandhane, Jessica Yung, Iuliya Beloshapka, Borja Ibarz, Vivek Veeriah, Lei Yu, Oliver Nash, Paul Lezeau, Salvatore Mercuri, Calle Snne, B. Mehta, Alex Davies, Daniel Zheng, F. Pedregosa, Yin Li, Ingrid von Glehn, M. Rowland, Samuel Albanie, Ameya Velingker, S. Schmitt, Edward Lockhart, Edward Hughes, H. Michalewski, Nicolas Sonnerat, D. Hassabis, P. Kohli, and David Silver. Olympiad-level formal mathematical reasoning with reinforcement learning. *Nature*, 651: 607 – 613, 2025.

- Paras Jain, Sam Kumar, Sarah Wooders, Shishir G. Patil, Joseph Gonzalez, and Ion Stoica. Skyplane: Optimizing transfer cost and throughput using cloud-aware overlays. *arXiv*, 2022.
- Pantea Karimi, Kimia Noorbakhsh, Mohammad Alizadeh, and Harinarayanan Balakrishnan. Improving coherence and persistence in agentic ai for system optimization. 2026.
- Lunsheng Li, Yang Chen, Hancheng Lu, Li He, Lei Gao, and Ningcheng Wang. Credit-r: Enhancing credit-based congestion control in cross-data center networks. In *2024 10th International Conference on Computer and Communications (ICCC)*, pages 1458–1463, 2024.
- A. Meyerson. Approximation algorithms for network design problems. *arXiv*, 2002.
- Alexander Novikov, Ngn Vu, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, Adam Zsolt Wagner, S. Shirobokov, Borislav M. Kozlovskii, Francisco J. R. Ruiz, Abbas Mehrabian, M. P. Kumar, Abigail See, Swarat Chaudhuri, George Holland, A. Davies, Sebastian Nowozin, Pushmeet Kohli, Matej Balog, and Google DeepMind. Alphaevolve: A coding agent for scientific and algorithmic discovery. *ArXiv*, abs/2506.13131, 2025.
- Venugopalan Ramasubramanian. Cost-aware resource management for decentralized internet services. *arXiv*, 2007.
- Trieu H. Trinh, Yuhuai Wu, Quoc V. Le, He He, and Thang Luong. Solving olympiad geometry without human demonstrations. *Nature*, 625:476 – 482, 2024.
- Chunhui Wan, X. Dai, Zhuo Wang, Minglei Li, Yanpeng Wang, Yinan Mao, Yushi Lan, and Zhiwen Xiao. Loongflow: Directed evolutionary search via a cognitive plan-execute-summarize paradigm. *ArXiv*, abs/2512.24077, 2025.
- Sarah Wooders, Shu Liu, Paras Jain, Xiangxi Mo, Joseph Gonzalez, Vincent Liu, and Ion Stoica. Cloudcast: High-throughput, cost-aware overlay multicast in the cloud. *arXiv*, 2024.

Reproducibility Statement

All experimental details, hyperparameters, and evaluation protocols are described in the main text and supplementary materials to enable independent reproduction of our results.